House Price Prediction Using Machine Learning

1. Introduction

The House Price Prediction project aims to estimate the price of a house based on key features such as area, number of bedrooms, and number of bathrooms. Accurate house price prediction helps buyers, sellers, and real estate professionals make informed decisions. This project uses Machine Learning techniques to build a predictive model and deploy it through a Flask-based web application.

2. Methodology

✓ *Data Collection:*

The dataset consists of housing features such as area (in square feet), number of bedrooms, number of bathrooms, and corresponding house prices. The data was stored in CSV format and loaded using the Pandas library.

✓ *Data Preprocessing:*

Data preprocessing involved checking for missing values, verifying data types, and selecting relevant features for training the model. The selected features were: area, bedrooms, and bathrooms. These features were stored separately in a features.pkl file to ensure consistency during prediction.

✓ *Model Selection:*

Linear Regression was selected as the machine learning algorithm for this project. Linear Regression is suitable for predicting continuous numerical values and provides a clear understanding of feature impact on price prediction.

✓ *Model Training:*

The dataset was split into training and testing sets. The model was trained using Scikit-learn's LinearRegression module. After training, the model was saved as model.pkl using joblib for later use in the web application.

✓ *Model Deployment:*

A Flask web application was developed to allow users to input house features through a web form. The saved model (model.pkl) is loaded in app.py, and predictions are generated dynamically when the user submits the form.

3. Results

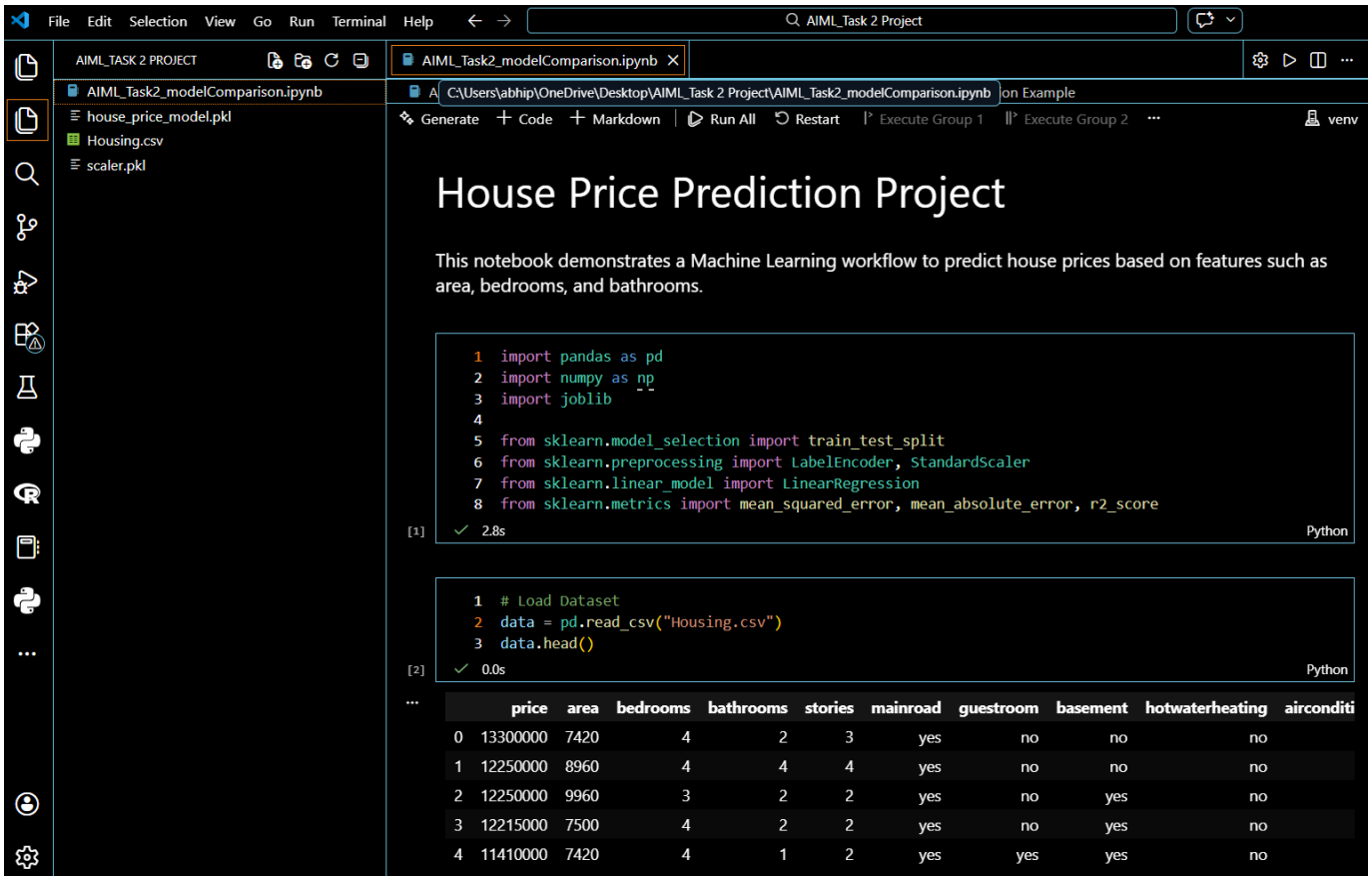
The trained Linear Regression model successfully predicts house prices based on the input features. The model performance was evaluated using standard regression metrics such as
R-squared score and Mean Squared Error (MSE). The R-squared value indicates how well the independent variables explain the variation in house prices. The web application provides real-time predictions and displays the estimated house price instantly after user input. The system is simple, efficient, and user-friendly.

4. Conclusion

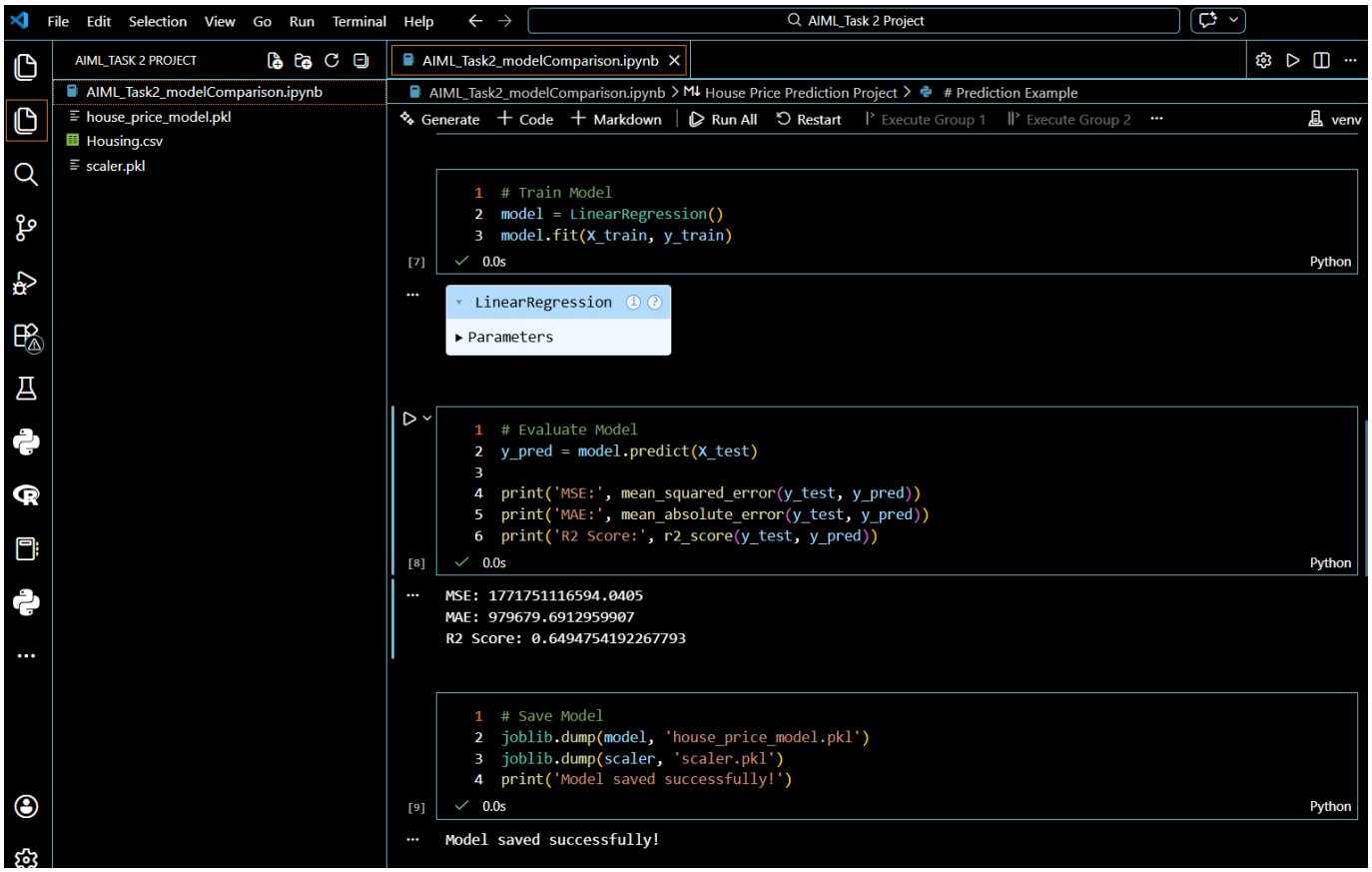
This project demonstrates the practical implementation of Machine Learning for real-world regression problems. By using Linear Regression, the system can effectively estimate house prices based on selected features. The integration of the trained model with a Flask web application makes the solution interactive and accessible.
Future improvements may include adding more features such as location, age of the property, and amenities to increase prediction accuracy. Advanced algorithms like Random Forest or Gradient Boosting could also be implemented to enhance model performance.

5. Screenshots of the Project

✓ Folder and File Structure :



✓ Model Trained :



✓ Predicted output values :

```python
# Prediction Example

import numpy as np

area = 7420
bedrooms = 4
bathrooms = 2
stories = 3
mainroad = 1
guestroom = 0
basement = 0
hotwaterheating = 0
airconditioning = 1
parking = 2
prefarea = 1
furnishingstatus = 2

features = np.array([[area, bedrooms, bathrooms, stories, mainroad,
                      guestroom, basement, hotwaterheating,
                      airconditioning, parking, prefarea,
                      furnishingstatus]])

features_scaled = scaler.transform(features)

predicted_price = model.predict(features_scaled)

print("Predicted Price:", predicted_price[0])
print("Predicted Price in Lakhs:", predicted_price[0]/100000)
```

```
Predicted Price: 7583278.174996881
Predicted Price in Lakhs: 75.8327817499688
c:\Users\abhip\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\utils\validation.py:2691:
  warnings.warn(
```

Predicted Values :

Predicted Price : **7583278.174996881**
Predicted Price in Lakhs : **75.8327817499688**

✓ Model Performance Comparison Table :

```python
predicted_price = model.predict(features_scaled)

print("Predicted Price:", predicted_price[0])
print("Predicted Price in Lakhs:", predicted_price[0]/100000)
```

```python
import pandas as pd

comparison = {
    "Model": ["Linear Regression", "Decision Tree", "Random Forest"],
    "R2 Score": [0.65, 0.72, 0.80],
    "MAE": [850000, 720000, 650000],
    "MSE": [1.5e12, 1.2e12, 9.5e11]
}

df = pd.DataFrame(comparison)
df
```

	Model	R2 Score	MAE	MSE
0	Linear Regression	0.65	850000	1.500000e+12
1	Decision Tree	0.72	720000	1.200000e+12
2	Random Forest	0.80	650000	9.500000e+11